TCT Station (BH552)

Henry Tan

06/24/2026

1 Overview

The TCT is designed to test the output of sensors on a circuit board and running IV curves. We will be using the infrared laser, so the instructions are designed for the use of the infrared laser, not the optical red laser.

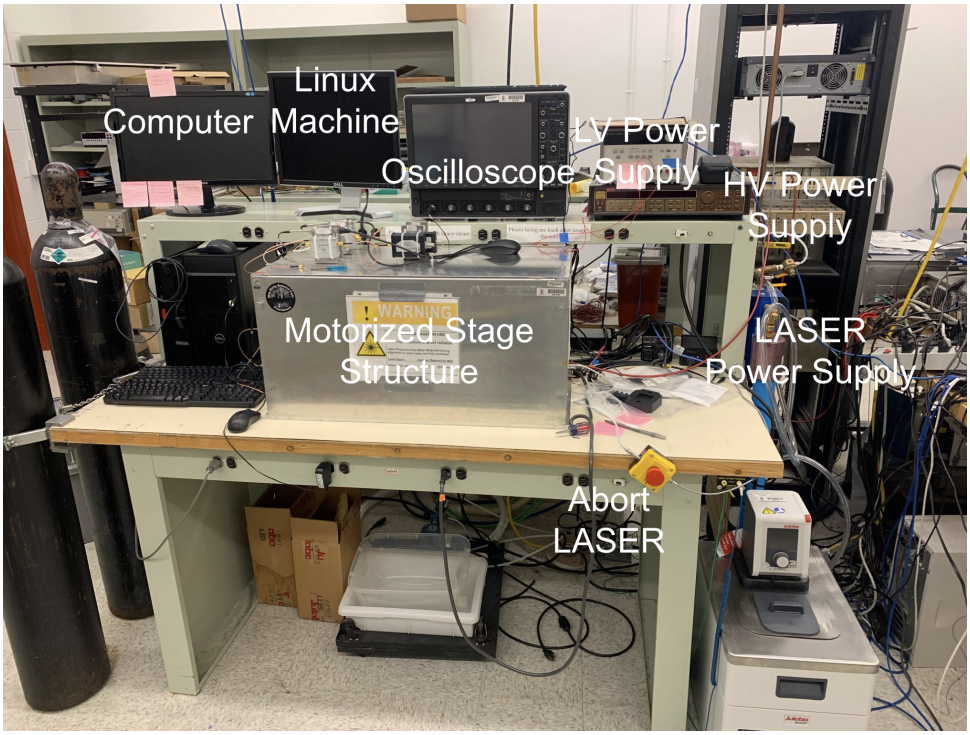


Figure 1: TCT setup in BH550. The leftmost monitor runs Windows and Labview. The rightmost monitor is the Linux where commands are executed. In the center is a LeCroy oscilloscope. The leftmost power supply (Agilent) provides the low voltage to the circuit board and the leftmost power supply (white Keithley) provides the bias / high voltage. Underneath the power supplies are the switches to turn on the LASER. Within the metal box contains the stage, holder for the circuit board, and the LASER itself. Finally, the red button at the edge of the desk is a button to stop the LASER.

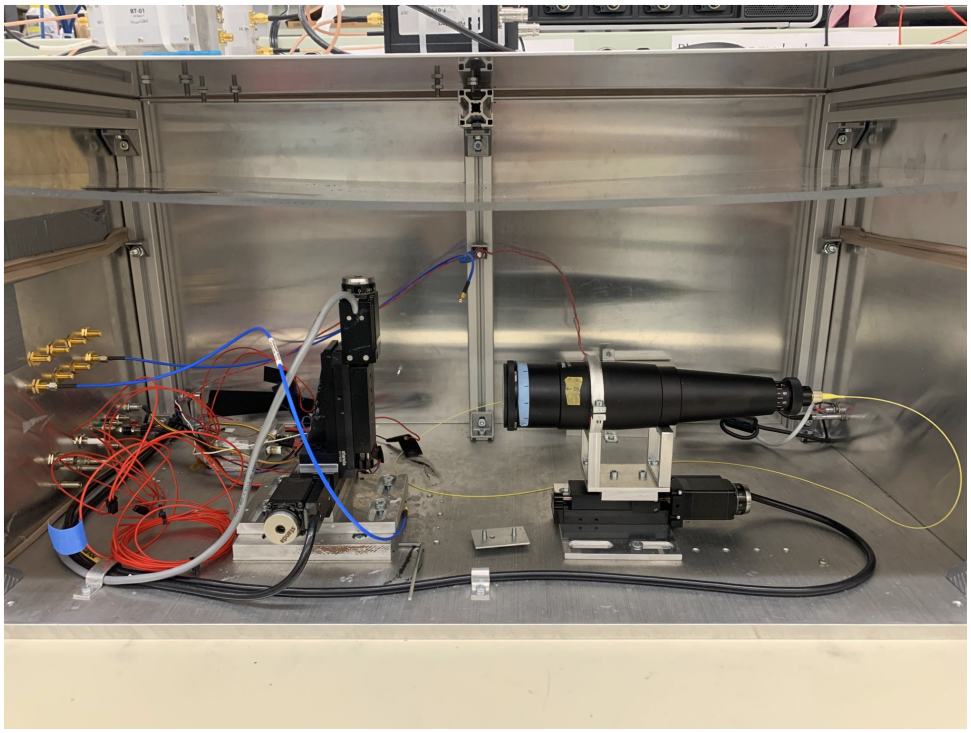


Figure 2: The inside of the TCT box. See Section 1 for an overview of the components.

2 IV Curves

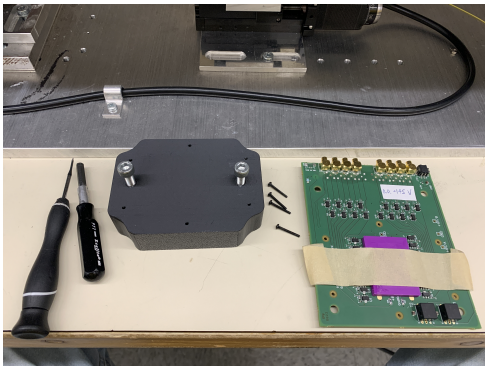
2.1 Setup

1. IV curves do not require operation of the oscilloscope, low voltage power supply, or LASER.
2. Attach the circuit board to the holder and mount

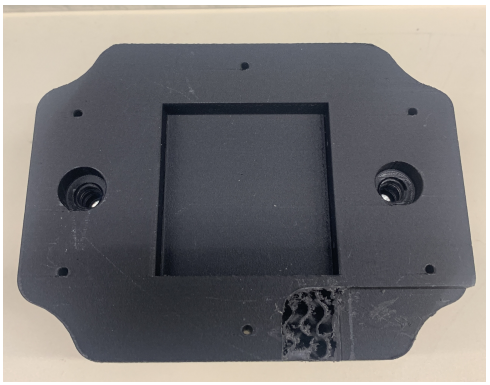
3 TCT Measurement

3.1 Setup

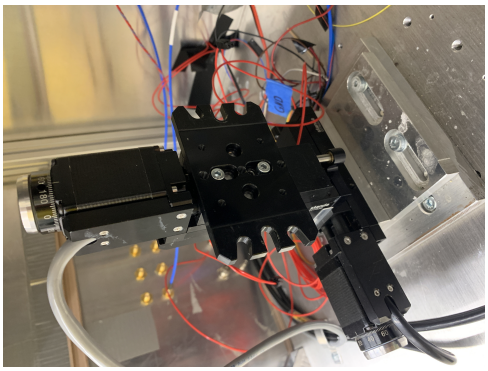
1. Open the front panel of the TCT box.
2. On the circuit board, determine which sensor you want to test and what output channel it is wired-bonded to and which SMA output the output channel leads to.
3. Mount the circuit board on the holder. Then attach the holder to the stage in the TCT.
4. Connect the blue cable to the SMA output that you have determined corresponds to the desired sensor. Check that the outer SMA cables (golden wires) connect to the oscilloscope. The recommended channel for the sensor output is Channel 1.
5. Attach the low voltage line (marked with blue tape) to the circuit board.
6. Take off the cap of the laser, rotating CCW, and the ND filter if applicable. Ensure the iris of the laser is completely open - the open direction is CW.
7. Turn on both the Windows computer and Linux machine. The login for the Windows is:
   (a) User: ".\LabUser"
   (b) Password: "SiLab2025"



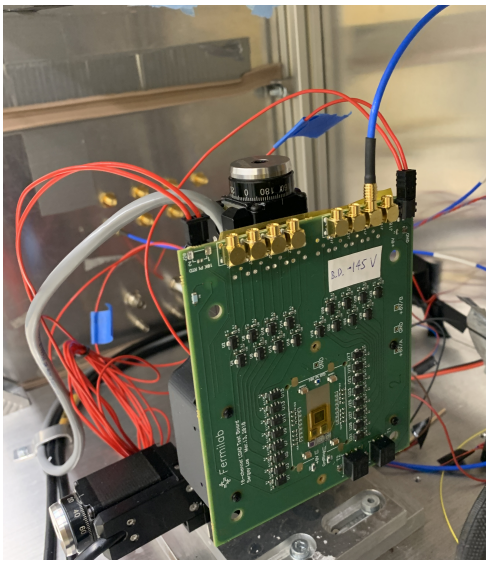
(a) 1a



(b) 1b



(c) 1c



(d) 1d

Figure 3

(c) Note: A singular mouse is shared between both computers. Unplug the mouse from one machine and switch to the other if necessary. The back keyboard corresponds to the Windows and front keyboard to the Linux.

8. Run the LabView script, remote_stage_controller, on Windows.
9. On the Linux machine, type "cd Desktop/TCT/RoloffLab".
10. Estimating the position of the sensor. If you have already finished the focusing procedure, then this step can be skipped.
    (a) WARNING: Before moving the stage, ensure that the wires are not in the way of the stage.
    (b) Use the black straight piece of plastic to estimate the position of the laser relative to the circuit board.
    (c) Type into the command line "movetoposition -e [axis] -d [distance]" to move the stage.
    (d) The parameter [axis] can take the argument "x, y, or z" and determines the axis at which the stage moves. [distance] determines the distance that the stage moves and can be negative. The stage moves in units of micrometers.
    (e) When the laser is roughly positioned on top of the sensor, type "./currentPositionR -s [First/.../Seventh]". This will save the position of the laser.
    (f) The parameter [First/.../Seventh] allows you to save multiple positions of the laser. Choose one from First to Seventh and stay with that argument throughout the data collection and analysis.
11. Reattach the cover of the TCT.
12. Turn on the oscilloscope, the laser, the LV power supply, and HV power supply.
13. Connect the oscilloscope to the network via an ethernet cable to the opposite wall.
14. Open Particulars LASER Controller on the Windows. Ensure that the USB status of the laser says "CONNECTED". Click on the tab called "Pulse width" and set the LASER pulse width to 30% and click "Enable DAC". Then switch to the tab "Singular frequency" and select 200 Hz. This will automatically turn on the laser. Check on the oscilloscope that the laser is on by looking at Channel 2. You should be able to observe the pulse triggers.
15. Set the LV voltage on the Agilent to 6.5V.

3.2 Focusing the Laser

1. This process is to ensure that the laser is directly aligned over the sensor. It is recommended to do the focusing procedure every time the holder is reinstalled, even if it is the same holder and circuit board to ensure the optimal position.
2. Type "sh run_focusRoloff.sh -m [compliance] -v [voltage] -a [axis]".
    (a) The flags are optional. Default compliance is set to 0.25mA, voltage to -100V, and axis to xyz.
    (b) The script will first bring the laser to the position of the sensor according to what was previously set.

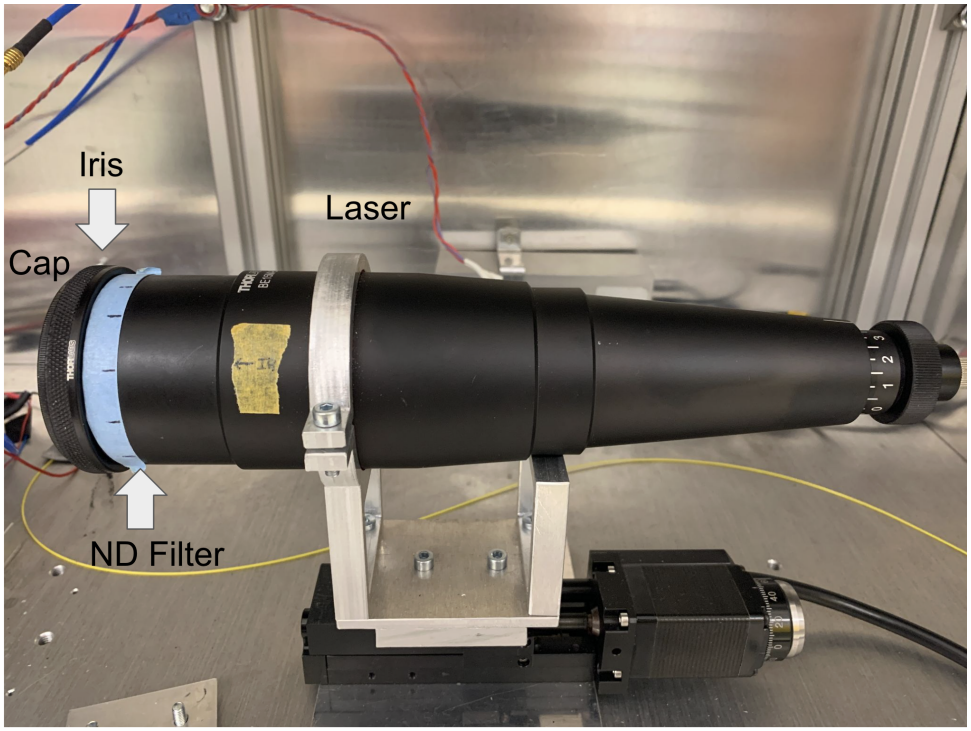


Figure 4: Diagram of the components on the laser. There are three attachments that are screwed onto the laser. The cap is an extremely thin piece of plastic that is the outermost layer. The iris has a noticeable bump compared to the rest of the laser that allows it to be turned without moving other components and the ND filter is lined by blue tape.

(c) It will then offset the laser and record an electronic noise reading. The file it outputs will be named "electronicNoise_First/.../Seventh_[x/y/z].bin".

(d) The laser will then move to the sensor at an offset position. The time out counter on the LabView script for the movement of the stage needs to hit 100 and an error will pop up on the Windows computer. Click "continue" and then the focus script will properly run.

(e) At the end of the sweeps, the greatest peak in sensor output will be determined and the position saved.

3. The script will output a png file called "[x/y/z]_[First/.../Seventh].png" that shows a plot of the integral of the sweeps vs the laser's relative position. Open this file by typing "xdg-open [x/y/z]_[First/.../Seventh]".

4. If the output of the plot is not as expected, either choose to rerun the focusing script at a different position (using movetoposition and currentPositionR) or re-running it at that position. You can also determine the ideal position and manually set the position of the laser to that point.

5. Repeat these steps for the other axis if the axis chosen was not "xyz"

6. When focusing is finished, delete the bin and png files by typing "rm *.bin" and "rm focus*.png".

3.3 Obtaining Data

1. When the position of the laser is determined, a proper measurement can be made.

2. Open the bash script to run the TCT by typing "sh runTCT_Roloff.sh -c [channel] -m [compliance] -n [device_name] -v [voltage] -i [initial voltage] -f [final voltage] -s [step size]".
   (a) The flags are optional.
   (b) The script will first take an electronic noise reading by offsetting the laser.
   (c) The TCT will then reposition back at the sensor and start sweeping from the initial voltage to the final voltage in the step size as set previously.
   (d) At the end, it will produce .bin files for each voltage it swept.
3. To analyze the bin files into readable pngs, use the python script "analyzeDataR.py". Execute the script by typing " /Desktop/TCT/RoloffLab/analyzeDataR.py -n [electronicNoiseBINfile] -f [nameOfSensor]*.bin -b n".
4. The results of the analyze data python script will output a png of each bin file that characterizes the signal evolution and a "results.txt" file that contains columns for the voltage, measured current, calculated pairs, and MIPS.
5. If you wish to plot pairs vs voltage, then type into the terminal " /Desktop/TCT/RoloffLab/plotResultsR.py results.txt". This will then output a "results.png" file that contains the plot (for the purposes of this group, this step will likely not be utilized).

A Bash / Unix Shell Commands

B Changing the Channel Output

NOTE: This documentation was made based on an original manual by Alex Buzzi on 06/14/25 for the same Probe Station setup in BH533.